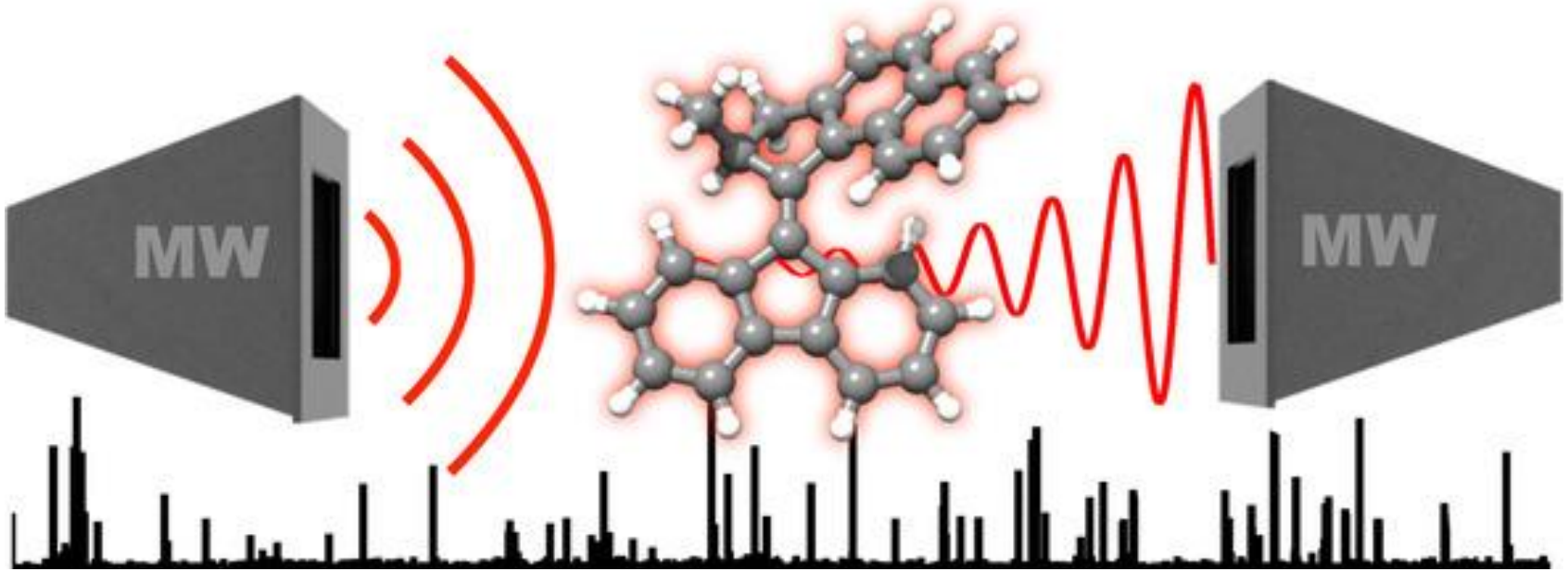
Rotational Spectroscopy

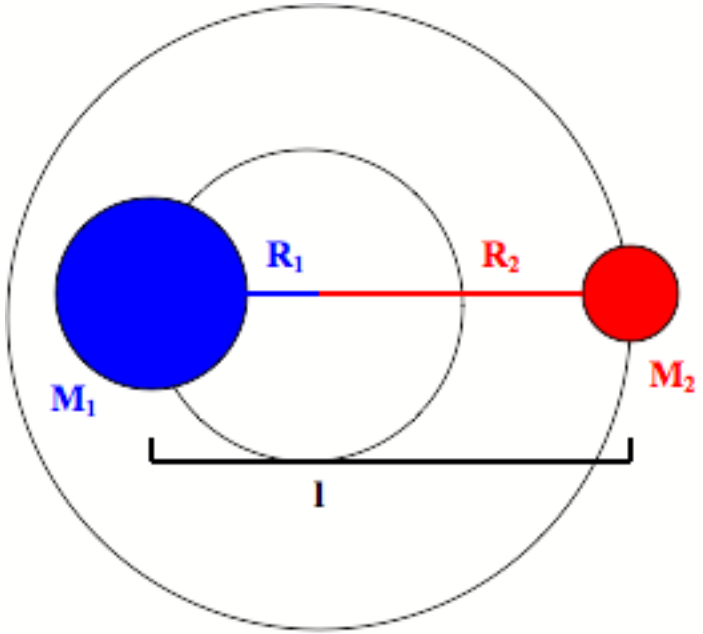


Angew. Chem. Int. Ed. 2017, 56, 11209-11212

Schrödinger Equation gives the rotational energy levels allowed to the rigid diatomic molecule.

$$E_J = \frac{h^2}{8\pi^2 I} J(J+1) = \frac{\hbar^2}{2I} J(J+1)$$ joules

$J = 0, 1, 2, \ldots\ldots$ ← **The rotational quantum number**



I is the moment of inertia, either I_A or I_B, since both are equal.

If the molecule absorbs a photon of frequency ν, the molecule can undergo a transition between rotational energy levels.

- What is the energy difference between the energy level with J+1 and the energy level with J?

$$\Delta E_{(J+1),J} = \frac{\hbar^2}{2I}(J+1)(J+2) - \frac{\hbar^2}{2I}J(J+1)$$

$$= \frac{\hbar^2}{2I}\left[J^2 + 3J + 2 - J^2 - J\right]$$

$$= \frac{\hbar^2}{2I}(2J+2) = \frac{\hbar^2}{I}(J+1)$$

Therefore, the frequencies at which transitions can occur are given by

$$\hbar\nu = \Delta E_{(J+1),J} = \frac{\hbar^2}{I}(J+1)$$

$$\nu = \frac{\hbar^2}{I\hbar}(J+1)$$

$$= \frac{\hbar^2}{2\pi I\hbar}(J+1)$$

$$= \frac{\hbar}{2\pi I}(J+1) = 2B(J+1)$$

The rotational constant **B** is given by: $B = \frac{\hbar}{4\pi I}$

In rotational spectroscopy, the spectra are usually discussed in terms of wavenumber.

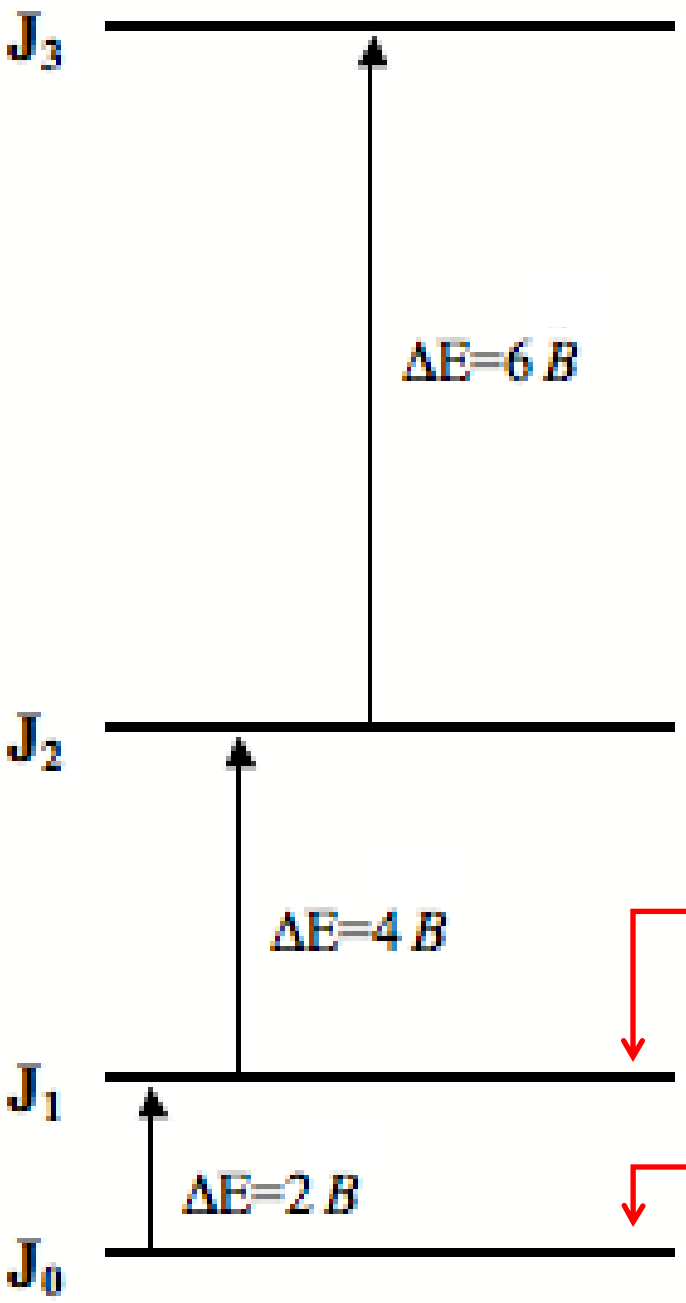
The energy in this unit is given as:

$$\varepsilon_J = \frac{E_J}{hc} = \frac{h}{8\pi^2 Ic} J(J+1)\,\mathrm{cm}^{-1} \qquad (\mathrm{J} = 0, 1, 2, ...)$$

$$\varepsilon_J = \frac{E_J}{hc} = BJ(J+1)\,\mathrm{cm}^{-1} \qquad (\mathrm{J} = 0, 1, 2, ...)$$

$$B = \frac{h}{8\pi^2 I_B c}\,\mathrm{cm}^{-1}$$

The allowed rotational energies of a rigid diatomic molecule.



For J=1, ε_J=2B

The rotating molecule has its lowest angular momentum.

For J=0, ε_J=0

The molecule is not rotating at all.

The Rotational Spectrum

Molecule at J=0 (Ground Rotational State)

→ No rotation occurs

The molecule can absorb the incident radiation to move to J = 1 state

The energy absorbed will be:

$\varepsilon_{J=1} - \varepsilon_{J=0} = 2B - 0 = 2B\ \text{cm}^{-1}$

$$\bar{\nu}_{J=0\rightarrow J=1} = 2B\,\text{cm}^{-1}$$

→ An absorption line will appear at 2B cm^{-1}.

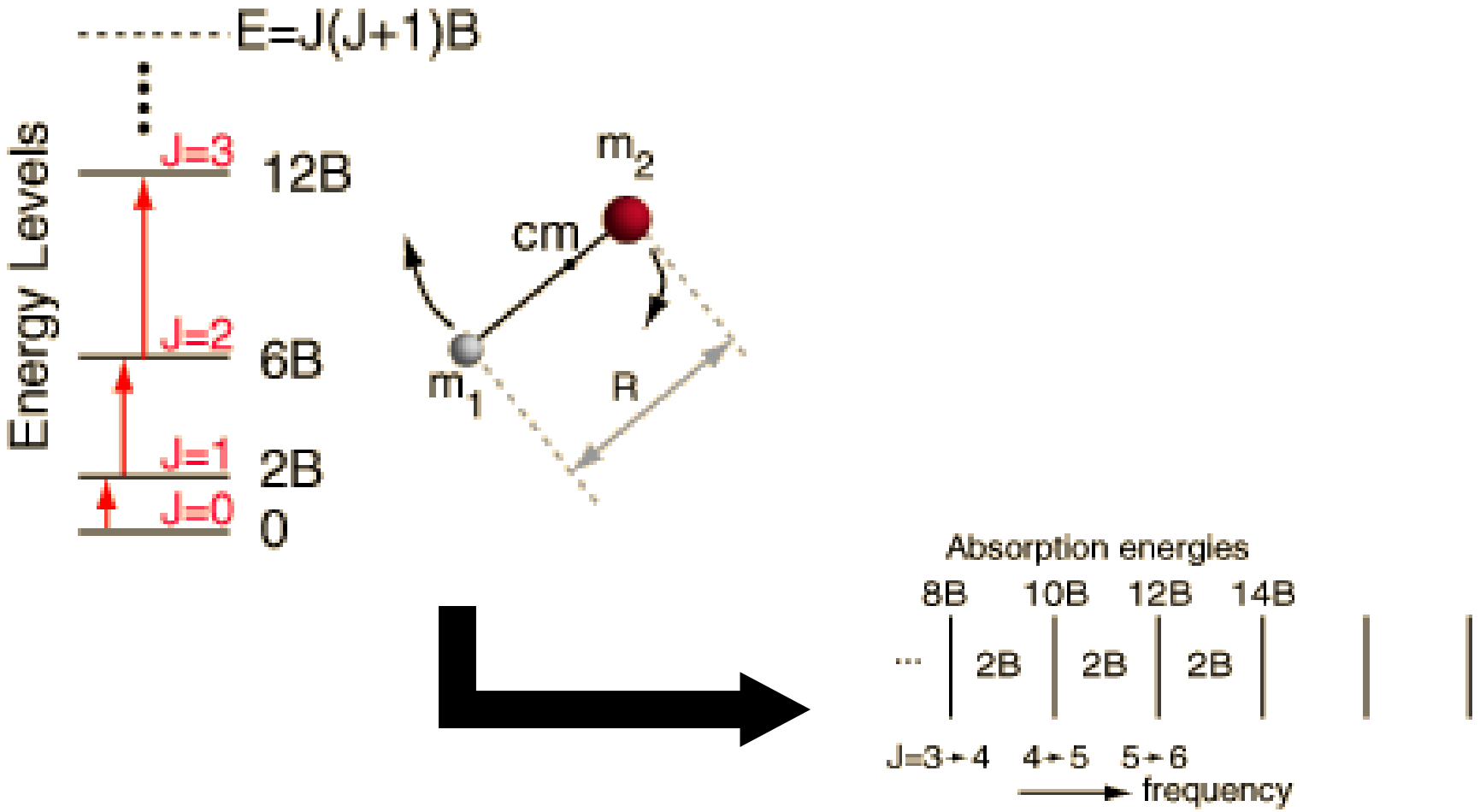
Molecule is raised from J=1 to J=2 by absorbing more energy.

$$\bar{\nu}_{J=1\to J=2} = \varepsilon_{J=2} - \varepsilon_{J=1}$$
$$= 6B - 2B = 4B\,\mathrm{cm}^{-1}$$

In general:
To raise the molecule from J to J+1

$$\bar{\nu}_{J\to J+1} = B(J+1)(J+2) - BJ(J+1)$$
$$= B\left[J^2 + 3J + 2 - \left(J^2 + J\right)\right]$$
$$= 2B(J+1)\,\mathrm{cm}^{-1}$$

Allowed transitions between the energy levels of a rigid diatomic molecules and the spectrum which arises from them.



To obtain the rotational spectrum for diatomic molecules:

Assumption:

Transition can take place from a particular level to its immediate neighbour, either above or below.

Application of Schrödinger wave equation shows that:

For the diatomic molecules we need to consider transitions **in which J changes by one unit.**

<u>Other transitions are spectroscopically forbidden</u>.

The Selection Rule

For rigid diatomic rotators:

$$\Delta J = \pm 1$$

The rotational spectrum is observed only if the molecule is asymmetric (heteronuclear), eg. HCl, CO.

The rotational spectrum is not observed if the molecule is homonuclear, eg. N_2, O_2.

HOMONUCLEAR DIATOMICS

Reasons for not observing a spectrum:

1. The rotation about the bond axis was rejected.
   The moment of inertia is very small about the bond.
   The energy levels will be extremely widely spaced.
   The molecule requires a huge amount of energy to be raised from J=0 to J=1 state.
   Such transitions do not occur under normal conditions.
   Result is no spectrum.

2. For such transitions there will be no dipole change.

HOMONUCLEAR DIATOMICS

Seven molecules do not show rotational spectrum.

HI BrONClF

Hydrogen
Iodine
Bromine
Oxygen
Nitrogen
Chlorine
Fluorine

The intensities of spectral lines

$$\bar{\nu}_{J \to J+1} = 2B(J+1)$$

Requirement:
A knowledge of relative probabilities of transition between the various energy levels.

Whether
a molecule has more or less chance of making the transition J=0 $\rightarrow$ J=1 than the transition J=1 $\rightarrow$ J=2.

Remember
ΔJ = $\pm$2, $\pm$3, etc. are forbidden (transition probability is zero).

The intrinsic probability that a single molecule in the J = 0 state will move to J = 1 is the same as that of a molecule moving from J = 1 to J = 2 in a gas sample.

But
there will be different number of molecules in each level.

Therefore, different total numbers of molecules will carry out transitions between various levels.

The line intensities are directly proportional to the initial numbers of molecules in each level.

The intensities of spectral lines

The three main factors:

1. Transition probability
2. Population of states
3. Path length of sample

Transition probability

Selection rules determine whether a transition from one state to another is probable or not.

Population of states

If there are N molecules distributed between two energy levels, E_{lower} and E_{upper},

At equilibrium: $$\frac{N_{upper}}{N_{lower}} = \exp(-\Delta E / kT)$$

Boltzmann distribution

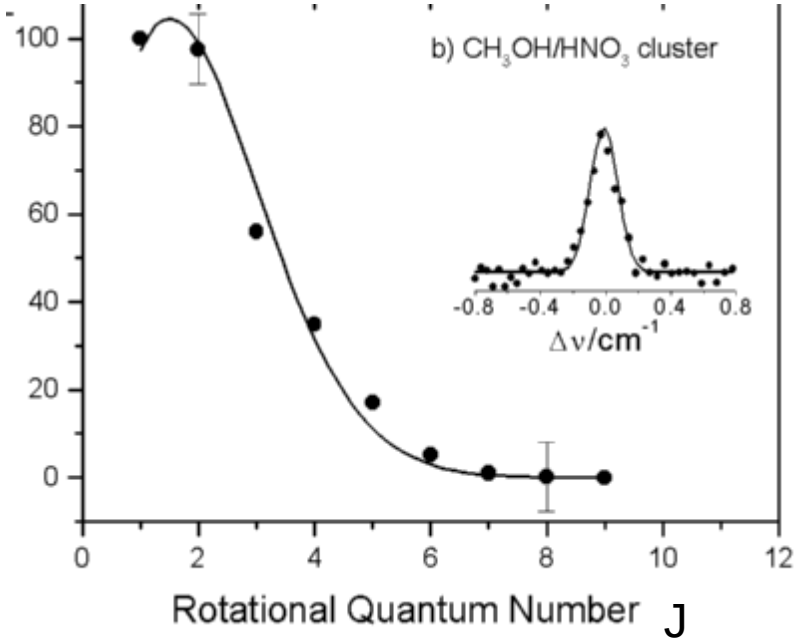
Rotational energy in the lowest level is zero, since J = 0.

If there are N_0 molecules present in this state then the content of any higher state is given by

$$\frac{N_J}{N_0} = \exp\left(-E_J/kT\right) = exp[-BhcJ(J+1)/kT]$$

N_J varies with J.

$exp[-BhcJ(J+1)/kT]$



Rotational state distributions of nascent OH fragments from 193 nm photolysis of HNO_3 in methanol clusters.